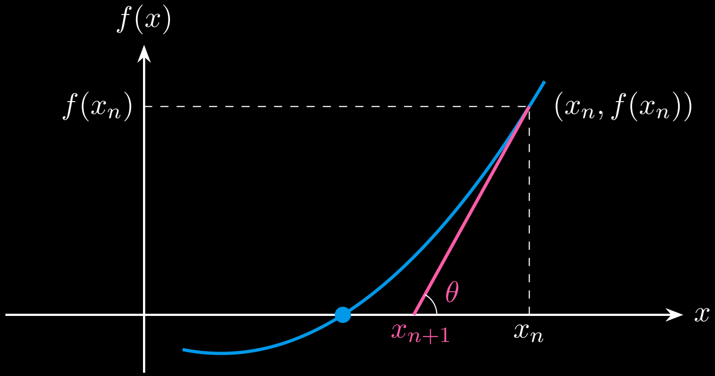
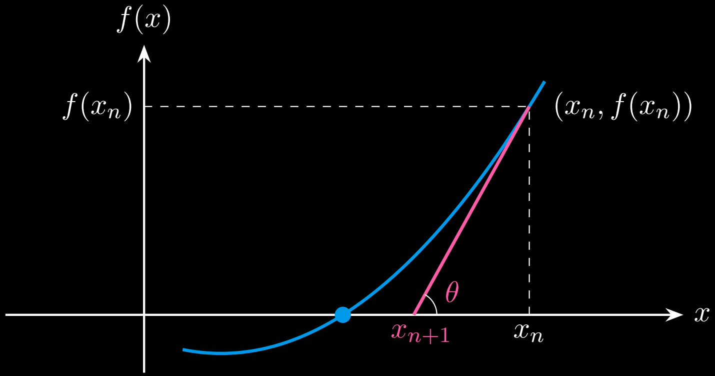
Intuitive Understanding of Newton's Method

$$\text{slope} = \frac{\text{change in } y}{\text{change in } x} = \frac{0 - f(x_n)}{x_{n+1} - x_n} = f'(x_n)$$



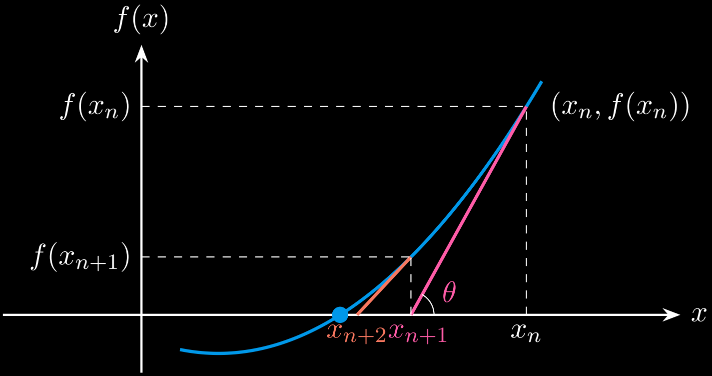
Intuitive Understanding of Newton's Method

$$x_{n+1} - x_n = -\frac{f(x_n)}{f'(x_n)} \quad \Rightarrow \quad x_{n+1} = x_n - \frac{f(x_n)}{f'(x_n)}$$



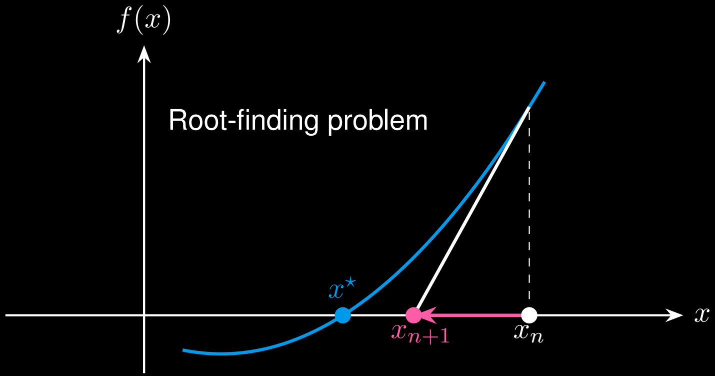
Intuitive Understanding of Newton's Method

$$x_{n+2} = x_{n+1} - \frac{f(x_{n+1})}{f'(x_{n+1})}$$



Intuitive Understanding of Newton's Method

$$x := x - \frac{f(x)}{f'(x)}$$ (update x until convergence)



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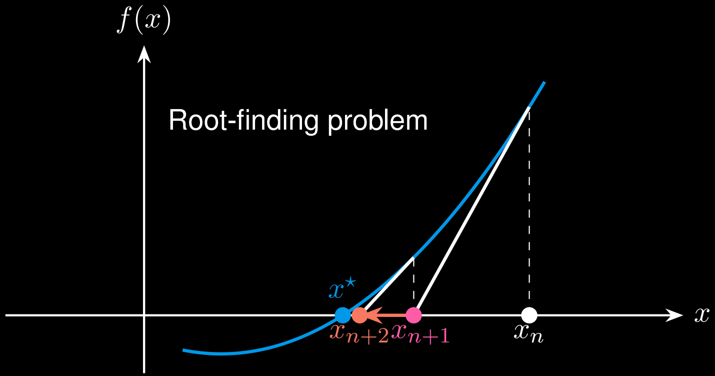

Unconstrained Nonlinear Optimization

Optimization problem:

$$\min_{x} \quad \underbrace{f(x)}_{\text{twice-differentiable}} \qquad \Rightarrow \qquad \text{Find} \quad f'(x) = 0$$

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Newton's method in optimization is a method to find the (local) minimum of a twice-differentiable function $f(x)$:

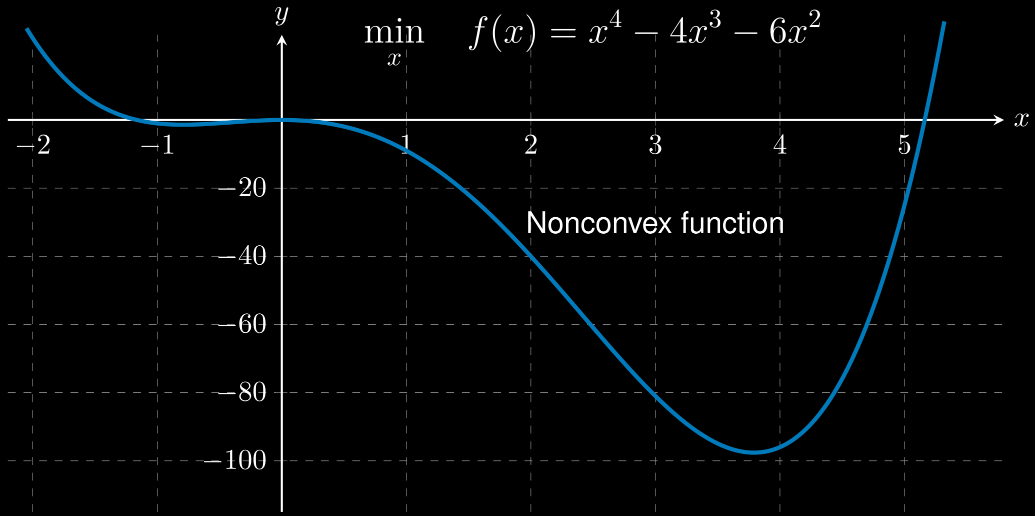
$$x := x - \frac{f'(x)}{f''(x)} \qquad \text{(update } x \text{ until convergence)}$$

with

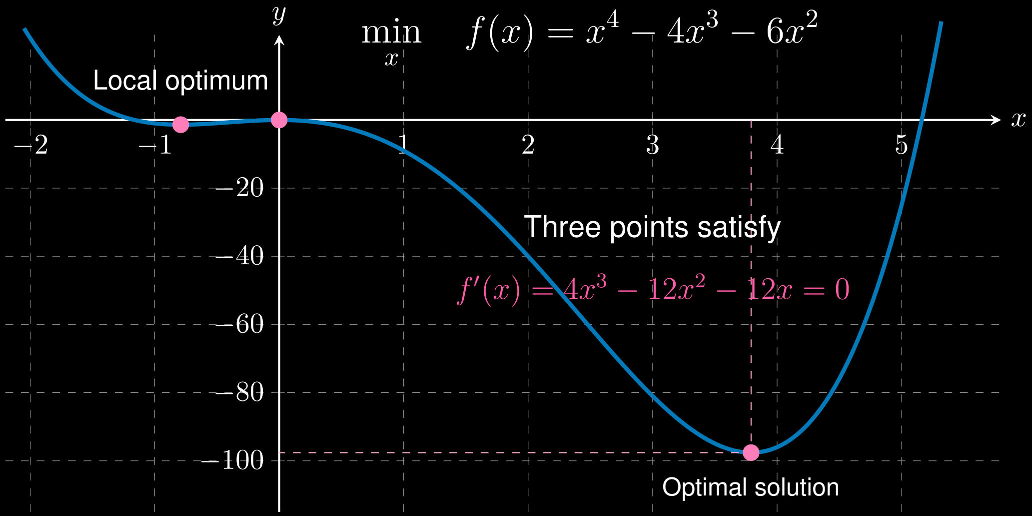
$f'(x)$ first-order derivative

$f''(x)$ second-order derivative

Unconstrained Nonlinear Optimization



Unconstrained Nonlinear Optimization



Unconstrained Nonlinear Optimization

Optimization problem:

$$\min_{x} \quad f(x) = x^4 - 4x^3 - 6x^2$$

Newton's method:

$$x := x - \frac{f'(x)}{f''(x)}$$ (update x until convergence)

with

$$f'(x) = 4x^3 - 12x^2 - 12x$$ and $$f''(x) = 12x^2 - 24x - 12$$

Unconstrained Nonlinear Optimization

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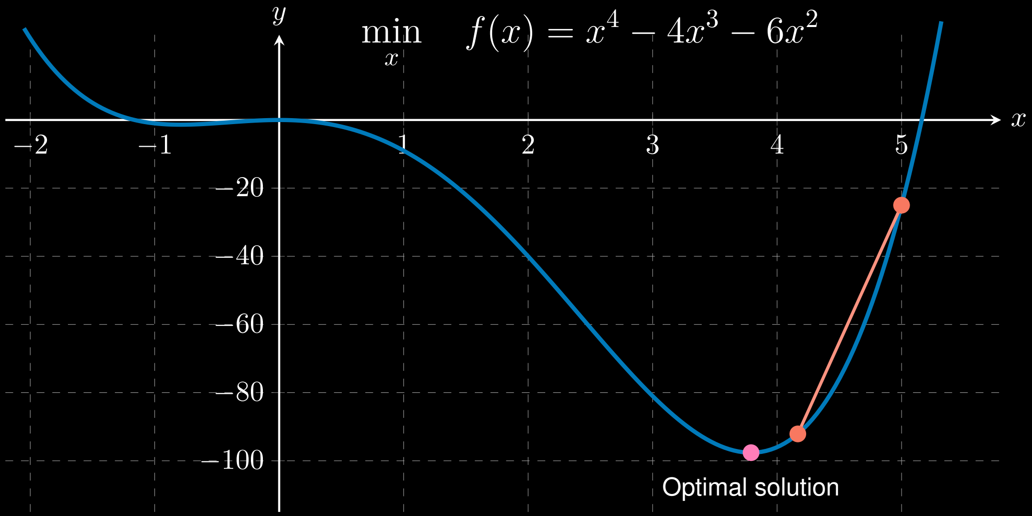
with

$$f'(x) = 4x^3 - 12x^2 - 12x \qquad \text{and} \qquad f''(x) = 12x^2 - 24x - 12$$

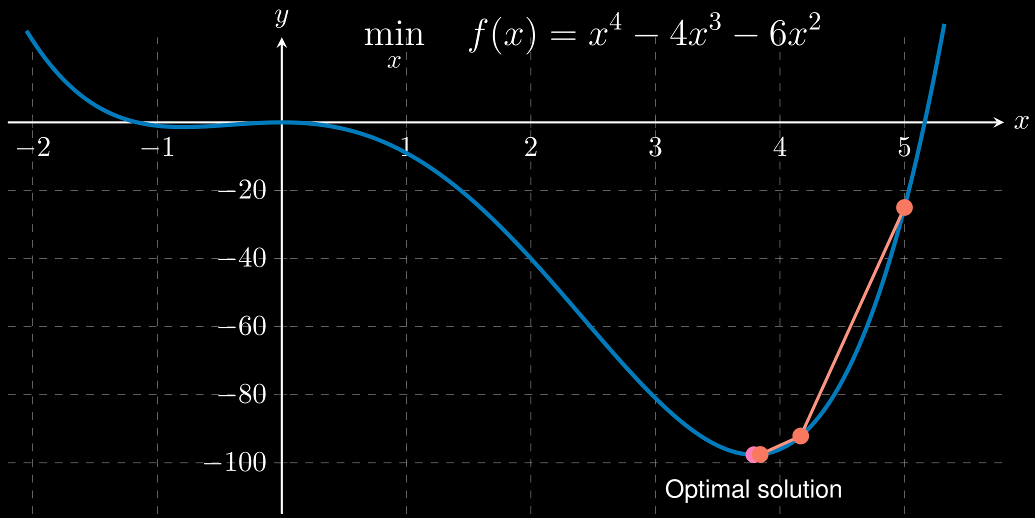
Given a starting point $x_0 = 5$:

$$f'(5) = 140, \quad f''(5) = 168 \qquad \Rightarrow \qquad x := 5 - \frac{f'(5)}{f''(5)} = 4.1667$$

Unconstrained Nonlinear Optimization



Unconstrained Nonlinear Optimization



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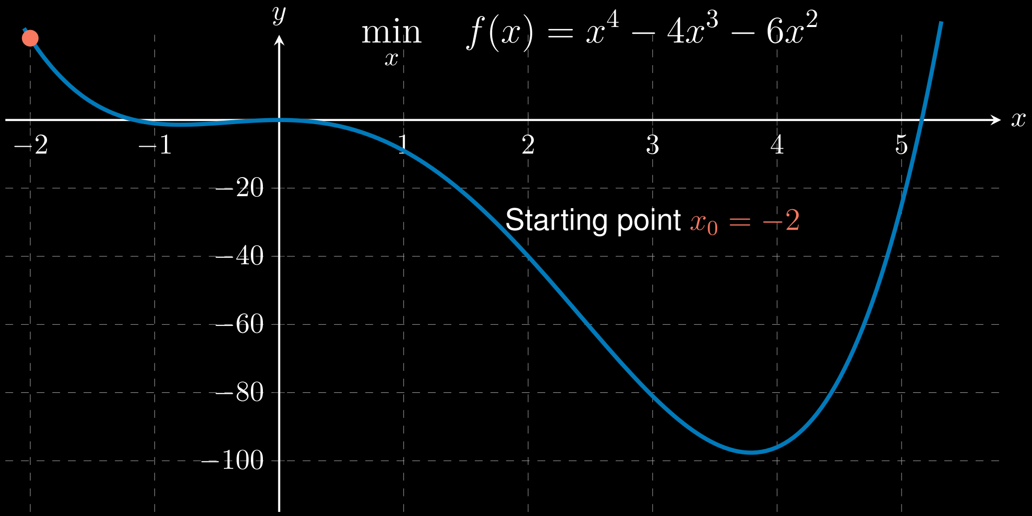

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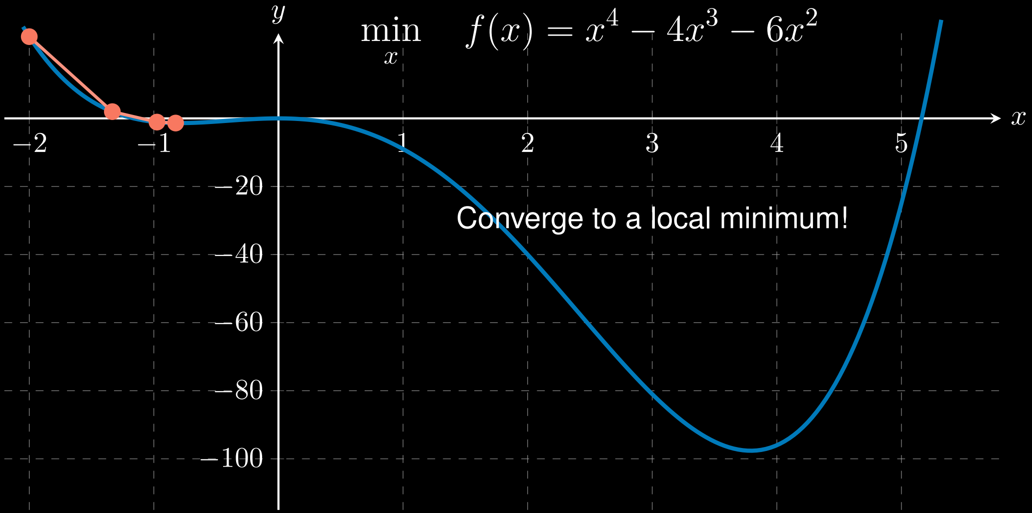
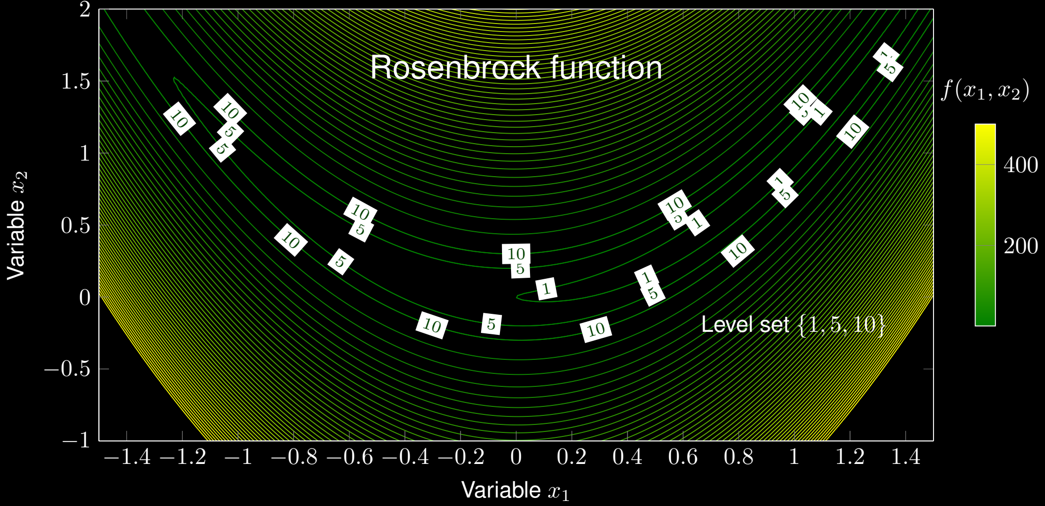


Converge to a local minimum!

Unconstrained Nonlinear Optimization

$$\min_{x_1, x_2} \quad f(x_1, x_2) = (1 - x_1)^2 + 100(x_2 - x_1^2)^2$$



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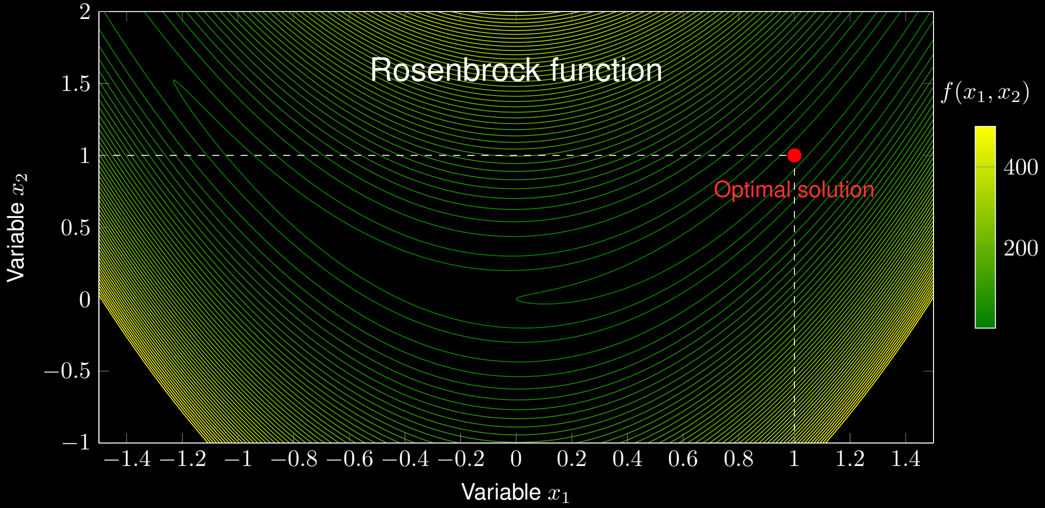

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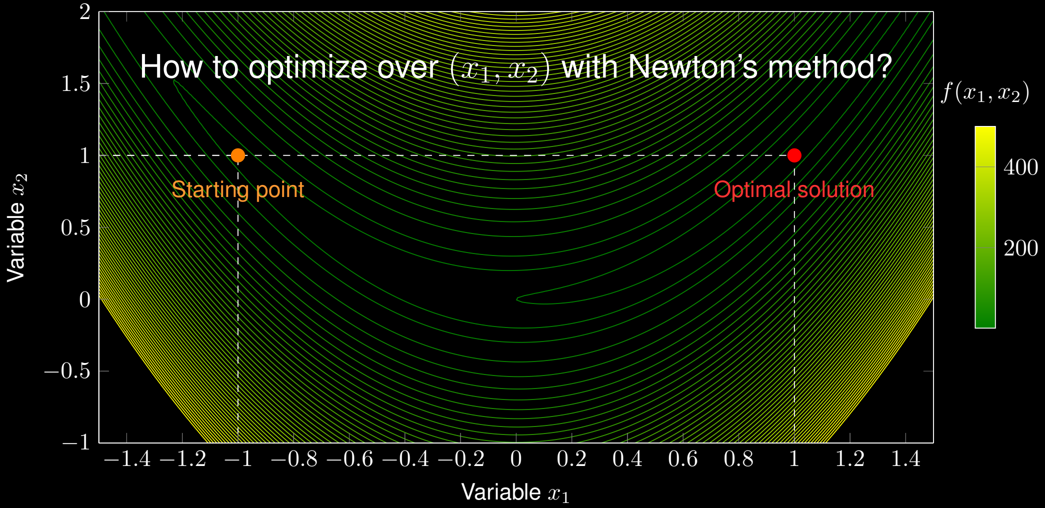

First- and Second-Order Derivatives

Optimization problem:

$$\min_{x_1, x_2} \quad \underbrace{f(x_1, x_2)}_{\text{twice-differentiable}} \quad = (1 - x_1)^2 + 100(x_2 - x_1^2)^2$$

First-order derivatives:

$$\frac{\partial f}{\partial x_1} = -2 + 2x_1 - 400x_1x_2 + 400x_1^3 \quad \text{and} \quad \frac{\partial f}{\partial x_2} = 200x_2 - 200x_1^2$$

First- and Second-Order Derivatives

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Second-order derivatives:

$$\frac{\partial^2 f}{\partial x_1^2} = \frac{\partial}{\partial x_1}\left(-2 + 2x_1 - 400x_1x_2 + 400x_1^3\right) = 2 - 400x_2 + 1200x_1^2$$

$$\frac{\partial^2 f}{\partial x_2^2} = \frac{\partial}{\partial x_2}\left(200x_2 - 200x_1^2\right) = 200$$

$$\frac{\partial^2 f}{\partial x_1 \partial x_2} = \frac{\partial^2 f}{\partial x_2 \partial x_1} = -400x_1$$

Unconstrained Nonlinear Optimization

The optimization can be converted into a root finding problem:

$$\min_{x_1, x_2} \quad f(x_1, x_2) \qquad \Rightarrow \qquad \text{Find} \quad \begin{bmatrix} \frac{\partial f}{\partial x_1} \\ \frac{\partial f}{\partial x_2} \end{bmatrix} = \mathbf{0} \quad \text{(first-order derivatives} = 0\text{)}$$

Newton's method for root finding:

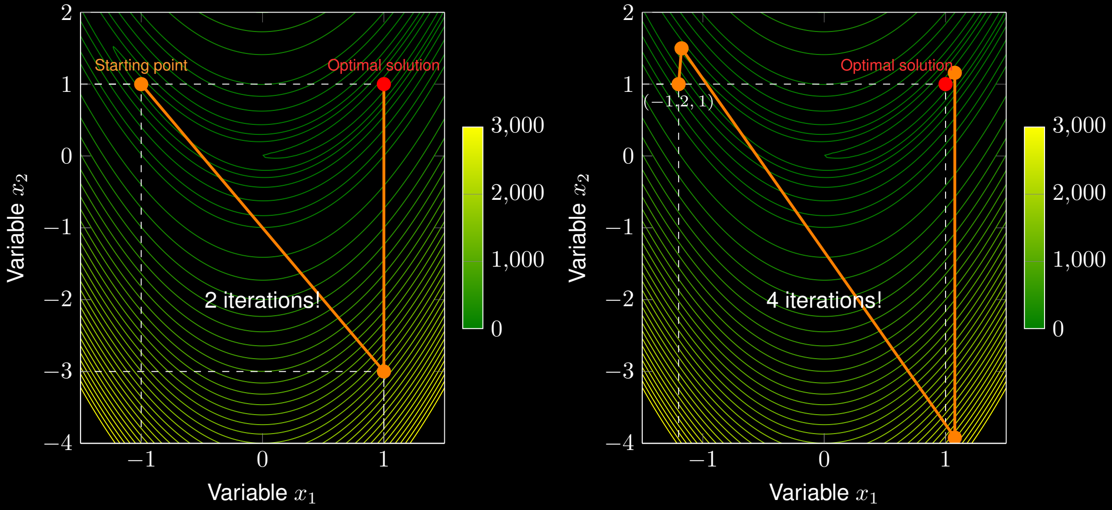
$$\begin{bmatrix} x_1 \\ x_2 \end{bmatrix} := \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} - \boldsymbol{H}^{-1} \begin{bmatrix} \frac{\partial f}{\partial x_1} \\ \frac{\partial f}{\partial x_2} \end{bmatrix}$$

with Hessian matrix

$$\boldsymbol{H} = \begin{bmatrix} \frac{\partial^2 f}{\partial x_1^2} & \frac{\partial^2 f}{\partial x_1 \partial x_2} \\ \frac{\partial^2 f}{\partial x_2 \partial x_1} & \frac{\partial^2 f}{\partial x_2^2} \end{bmatrix} = \begin{bmatrix} 2 - 400x_2 + 1200x_1^2 & -400x_1 \\ -400x_1 & 200 \end{bmatrix}$$

Newton's Method in Unconstrained Nonlinear Optimization

$$\min_{x_1, x_2} \quad f(x_1, x_2) = (1 - x_1)^2 + 100(x_2 - x_1^2)^2$$



Thanks for your attention!

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